

Yirui (Ray) Wang

Phone: (858) 396-4999 | Email: rayyiruiwang@gmail.com | [LinkedIn](#) | [GitHub](#)

EDUCATION

University of California, San Diego

La Jolla, California

Bachelor of Science in Mathematics and Computer Science

Sep 2024 – May 2028

- **GPA: 3.95 / 4.0**
- **Relevant Coursework:** Linear Algebra, Calculus, Introduction to Data Science, Combinatorics, Object-Oriented Programming, Data Structures & Algorithms, Advanced Data Structures, Computer Organization, Systems Programming

TECHNICAL PROJECTS

AI Interviewer System | Python, FFmpeg, LLM, LLaMA-based evaluation

Feb 2026 – Feb 2026

- Built an AI coding interviewer using the Grok LLM API to conduct technical interviews based on job descriptions
- Implemented speech-to-text input and timed responses to analyze answers and adapt follow-up questions dynamically
- Developed answer summary sheet, video recording support, and revision suggestions to help users improve performance

Just Walk Out System | Python, Pytorch, Loss Function, IoU

Oct 2025 – Jan 2026

- Built a computer vision prototype to enable checkout-free shopping through automatic product recognition
- Studied YOLO architecture and reproduced IoU and loss functions using Python and PyTorch on Google Colab
- Implemented and validated a fruit-detection pipeline that correctly identified customer-selected items in test scenarios

Save Our Seconds (SOS) | Python, API, SQL, FastAPI

Oct 2025 – Jan 2026

- Designed a mobile web supporting heart-attack emergencies to support bystanders to adopt first aid treatment efficiently
- Integrated CPR video tutorial and AED location maps via Google Maps API, serving backend data with FastAPI and SQL
- Delivered a live demo to 60+ attendees and judges, winning the Audience Popularity Prize among 9 competing teams

RESEARCH EXPERIENCE

Devices Monitoring System Development | Python, Numpy, Matplotlib, Pandas, Linux

Jul 2025 – Oct 2025

Undergraduate Research Assistant

Shanghai, China

- Applied electromagnetic and acceleration signal analysis to characterize device behavior and identify operating states
- Used existing Python scripts on Linux to collect and process electromagnetic signals into PSD spectra, analyzed frequency-domain patterns, and contributed to 20% core device state classification judgment logic and scripts
- Enabled real-time device state monitoring and visualization by integrating judgment results with a frontend interface

EXTRACURRICULAR ACTIVITIES

Association of Computing Machinery (ACM)

Jan 2025 – Present

- Employed the JavaScript MERN stack, including MongoDB, Express.js, React.js, and Node.js to build a poll website
- Automated structured data extraction from Wikipedia's Hypertext page, gaining practical web scraping experience
- Explored diffusion models including forward noise injection and denoising sampling mechanisms

SKILLS

- **Programming Languages:** Python, Java, C, C++
- **Technical Skills:** Data Analysis (Pandas & NumPy), SQL, Pytorch
- **Machine Learning:** Supervised Learning (Regression & Classification), Computer Vision, Model Evaluation
- **Development Tools:** Git, Github, Linux, Google Colab, JupyterNotebook, VS Code, MATLAB